

Learning on topological surface and geometric structure for 3D molecular generation

Received: 28 February 2023

Accepted: 6 September 2023

Published online: 9 October 2023

 Check for updates

Odin Zhang^{1,4}, Tianyue Wang^{1,4}, Gaoqi Weng¹, Dejun Jiang¹, Ning Wang², Xiaorui Wang², Huifeng Zhao¹, Jialu Wu¹, Ercheng Wang³, Guangyong Chen³, Yafeng Deng², Peichen Pan¹, Yu Kang¹✉, Chang-Yu Hsieh¹✉ & Tingjun Hou¹✉

Highly effective de novo design is a grand challenge of computer-aided drug discovery. Practical structure-specific three-dimensional molecule generations have started to emerge in recent years, but most approaches treat the target structure as a conditional input to bias the molecule generation and do not fully learn the detailed atomic interactions that govern the molecular conformation and stability of the binding complexes. The omission of these fine details leads to many models having difficulty in outputting reasonable molecules for a variety of therapeutic targets. Here, to address this challenge, we formulate a model, called SurfGen, that designs molecules in a fashion closely resembling the figurative key-and-lock principle. SurfGen comprises two equivariant neural networks, Geodesic-GNN and Geoatom-GNN, which capture the topological interactions on the pocket surface and the spatial interaction between ligand atoms and surface nodes, respectively. SurfGen outperforms other methods in a number of benchmarks, and its high sensitivity on the pocket structures enables an effective generative-model-based solution to the thorny issue of mutation-induced drug resistance.

Rational drug design is generally performed via two approaches: ligand-based drug design and structure-based drug design (SBDD)¹. SBDD is widely believed to be more effective in delivering novel drug candidates², as it models how proteins interact with small molecules to induce the ensuing biological effects. SBDD offers two main protocols: virtual screening³, where the affinity between the molecules in a compound library and a protein target is evaluated and the top-ranked molecules are selected for subsequent experimental validation, and molecular generation², where compounds are designed to bind strongly to a protein by optimizing the atomic interactions in the pocket. Combinatorial chemistry has revealed that the number of potential drug-like molecules can be as large as 10^{60} , and the combination of all compound

libraries is still a drop in the ocean in comparison. Virtual screening is a search for lead compounds in the midst of such a highly confined chemical space, which is a major reason for the high false-positive rate of virtual screening. In contrast, structure-based molecular design does not presuppose a predefined chemical space, but rather creates candidate molecules automatically by assembling some elementary constituents (such as atoms or molecular fragments) under the guidance of a fitness function. LUDI⁴ and LigBuilder⁵ are representatives of traditional approaches. However, the scoring functions they utilize have proven to be of limited accuracy for virtual screening⁶. Thus, they cannot efficiently produce promising compounds as the molecular generation process is guided by a biased scoring function, which may

¹Innovation Institute for Artificial Intelligence in Medicine of Zhejiang University, College of Pharmaceutical Sciences, Zhejiang University, Hangzhou, China. ²Hangzhou Carbonsilicon AI Technology Co., Ltd, Hangzhou, China. ³Zhejiang Lab, Zhejiang University, Hangzhou, China. ⁴These authors contributed equally: Odin Zhang, Tianyue Wang. ✉e-mail: yukang@zju.edu.cn; kimhsieh@zju.edu.cn; tingjunhou@zju.edu.cn

drive the generation to deviate from where some real bioactive compounds are located in the chemical space. Therefore, there is often no practical advantage for many existing generative models over virtual screening even if the generation process can theoretically access far more regions in the chemical space.

The study of interactions between protein pockets and small molecules is key to addressing these challenges. According to a famous theory, they can be abstracted as a lock-and-key model⁷, as shown in Supplementary Fig. 1a, emphasizing a match between the pair for two conditions, that is, a geometric match in the form of shape complementarity (the recessed part of the lock hole fits nicely with the protruding part of the key) and an energetic match, such as the mutual attraction of positive and negative charges. In principle, a successful SBDD can be achieved based on an accurate evaluation of these two factors. Conventionally, interactions between pockets and molecules are modeled by decomposing the overall energy function into different contributions based on physical reasoning, and most previous machine learning methods have determined the parameters of a model by fitting the weights of the energy terms⁸. The recent rise of artificial intelligence for science offers a fresh perspective on how to model molecular interactions and provides a highly flexible computational framework to derive a data-driven characterization of pocket–molecule interactions. For instance, convolutional neural network-based scoring functions⁹ are able to automatically capture the spatial features of the distribution of small molecules within protein pockets, and graph neural network-based scoring functions^{10,11} are able to process graph structure data of proteins and small molecules by taking into account the topological relationships between them to some extent. Another approach, represented by PIGNet¹², restricts the space of interactive features to the space of physically meaningful interactions and performs nonlinear transformations afterward to enhance the model's predictive power. Recently, a new characterization approach¹³ has emerged in the field of protein–protein interaction (PPI) prediction, where proteins are modeled as abstract surfaces. This approach takes into account the compatibility of protein–protein interface geometry and is able to capture intricate interface–interface interactions more directly due to the explicit modeling of protein topology.

Encouraged by the impressive performance of this surface representation for modeling PPI¹³, we adopt the protein surface channels as the protein representation for SBDD, embed relevant energy features on the surface and then perform the topological learning via our designed Geodesic-Graph Neural Network (GNN) framework. Unlike the cases for PPI predictions, the binding interactions of small molecules within protein pockets are dominated by spatial interactions, such as van der Waals force and electrostatic force, and therefore we propose another framework called Geoattn-GNN to learn this additional and crucial contribution of binding interactions. Our proposed model, SurfGen, which unifies both frameworks, can not only learn such geometric interactions but also better model the complementarity between small molecules and protein pockets through topological learning.

In addition to learning favorable interaction patterns for binding, structure-based molecular generation needs to specify the atomic coordinates of the proposed compound within a protein pocket. To do this reliably, it is crucial to properly embed the physics symmetry into the network design by making

$$R \circ f(x) = f(R \circ x) \quad (1)$$

or

$$f(x) = f(R \circ x) \quad (2)$$

where R is the rotation matrix, $\circ$ is the matrix multiplication, and $f(x)$ is the output of the neural network f of a given input x . Equation (1) refers

to the equivariant property, while equation (2) denotes the invariant property. By combining recently developed techniques¹⁴ and the nature of the lock-and-key model, we design the Geodesic-GNN and Geoattn-GNN blocks for enforcing equivariance in SurfGen. Further discussions and the sketch of proof can be found in Supplementary Section 2.

In summary, inspired by the figurative analogy of the celebrated lock-and-key model, we develop a machine learning solution, SurfGen, for structure-based molecule design. SurfGen's two key components, Geodesic-GNN and Geoattn-GNN blocks, are constructed to learn topological and geometric interactions, respectively. The learning of these molecular interactions helps to capture the appropriate topological and spatial distribution patterns for the atoms within a target pocket. We conduct extensive experiments to prove that the molecules generated by SurfGen not only possess higher binding affinity to the protein pockets compared with the original ligands and those generated by other state-of-the-art (SOTA) approaches but also form optimal geometric and energetic complementarities with the protein pockets. In addition, the results on real-world drug discovery cases demonstrate that SurfGen-generated molecules show the highest similarity to the experimental active molecules, and exemplary molecule generation has been done to expand a focused compound library targeting coronavirus disease (COVID-19). Moreover, the experiments on mutated targets demonstrate that SurfGen is sensitive enough to perceive the implications of critical mutations of a protein target, and noticeable mutation-induced impacts on the generated molecules can be observed.

Results

As an SBDD approach, SurfGen is supposed to evaluate the affinity between the generated molecule and the pocket. Most molecule-generation models are ligand centric¹⁵, and they are primarily used to explore the chemical space, so the assessments in many earlier studies focused more on the intrinsic molecular properties, such as drug-like properties, synthesizability, diversity and novelty. In contrast, SurfGen and other recently proposed SBDD approaches^{16–18} aim for a more specific goal of generating compounds that can bind tightly to a given protein target. Hence, for these newer models, the binding affinity of generated molecules should be used as a fundamental metric for model evaluation. To this end, we have collected some important therapeutic targets and the associated experimentally validated molecules to evaluate whether these models including SurfGen could generate competitive drug candidates in real-world drug discovery projects.

The baselines used in this study cover a broad range of models, from the recently developed deep learning-based GraphBP¹⁸ and Pocket2Mol¹⁶, to the reinforcement learning-based Morld¹⁹ and the traditional method, LigBuilder⁵. Both GraphBP and Pocket2Mol directly generate molecules in an end-to-end manner, whereas GraphBP predicts the internal coordinates of molecules and Pocket2Mol predicts the relative coordinates. Morld, a reinforcement learning-based method, depends on the docking software, AutoDock Vina²⁰, to construct the rewards for guiding molecular generation; and Ligbuilder utilizes their own binding energy prediction method, SCORE,²¹ and a genetic algorithm to control the construction of a ligand by iteratively adding fragments.

Molecular generation on test set

The dataset used to train SurfGen is CrossDock2020²², which enriches the training pool for the protein–ligand structure data with molecular docking. Autodock Vina²⁰ is used here to assess the affinity between small molecules and protein pockets. It is worth noting that most other methods^{16,18,23} also employ Vina for affinity assessment, but they often perform the docking mode, that is, estimating the protein–ligand binding energy after the conformational search within the software. Although this docking protocol truly assesses the binding strength of

Table 1 | The mean binding energies and drug-like properties for the top-1, -3, -5 and -10 molecules

	Test Set	GraphBP	Pocket2Mol	World	LigBuilder	SurfGen
Top 1						
Docking energy (↓)	-7.158	-9.332	-9.247	-8.272	-10.057	-10.461
Scoring energy (↓)	-5.998	0.082	-8.061	*	-8.367	-8.727
Hit pocket (%)	–	87.07	92.10	72.00	94.68	98.68
QED	0.484	0.560	0.562	0.076	0.603	0.541
SA	0.732	0.468	0.707	0.343	0.674	0.609
Lipinski	4.367	4.821	4.921	3.160	4.929	4.877
LogP	0.947	1.552	0.8249	6.646	0.885	0.624
Top 3						
Docking energy (↓)	-7.158	-8.809	-9.042	-7.866	-9.750	-10.254
Scoring energy (↓)	-5.998	3.812	-7.850	*	-8.017	-8.476
Hit pocket (%)	–	79.91	88.59	72.00	94.68	98.68
QED	0.484	0.567	0.574	0.069	0.605	0.528
SA	0.732	0.474	0.712	0.339	0.690	0.628
Lipinski	4.367	4.787	4.938	3.093	4.921	4.766
LogP	0.947	1.470	0.786	6.748	0.869	0.592
Top 5						
Docking energy (↓)	-7.158	-8.515	-8.924	-7.623	-9.565	-10.165
Scoring energy (↓)	-5.998	7.182	-7.708	*	-7.799	-8.334
Hit pocket (%)	–	75.05	86.57	72.00	94.68	98.42
QED	0.484	0.563	0.572	0.069	0.615	0.530
SA	0.732	0.478	0.719	0.342	0.686	0.622
Lipinski	4.367	4.776	4.928	3.104	4.927	4.757
LogP	0.947	1.430	0.778	6.826	0.869	0.689
Top 10						
Docking energy (↓)	-7.158	-8.060	-8.730	-7.218	-9.273	-9.959
Scoring energy (↓)	-5.998	15.566	-7.449	*	-7.478	-8.085
Hit pocket (%)	–	64.51	84.21%	72.00	94.60	97.89
QED	0.484	0.569	0.575	0.062	0.631	0.538
SA	0.732	0.485	0.721	0.337	0.696	0.638
Lipinski	4.367	4.778	4.944	2.989	4.939	4.824
LogP	0.947	1.366	0.787	6.987	0.805	0.663

The asterisks indicate that the model cannot be tested on this score, while the dashes signify that molecules in the test set have no HitPocket metric since this metric is defined on them. The docking energy and scoring energy are computed from Autodock Vina²⁰. Hit pocket represents the probability of the targets that have the generated molecules with higher binding affinity than their original ligands. QED⁴⁶ denotes the quantitative estimation of drug likeness, an integrative score that evaluates compounds' favorability to become a hit. SA⁴⁷ is the synthetic accessibility evaluation, a score that estimates the difficulty of synthesizing a compound—a higher score corresponds to a more easily synthesized molecule. Lipinski⁴⁸ is Lipinski's rule-of-five, a rule of thumb to evaluate drug likeness—the higher the better. logP⁴⁹ is the octanol–water partition coefficient, and logP values should generally be between -0.4 and 5.6, and ideally between 1.35 and 1.8 for good oral intestinal absorption.

small molecules in the pocket, it is crucial to consider that the three-dimensional (3D) pocket molecular generative model uses acquired knowledge to directly position within the pocket. Consequently, the additional uncertainty could be introduced by the conformational search of the software, as the proposed poses might substantially deviate from the conformations the model initially generates. As a solution, the in situ estimation of binding energy—scoring energy—is introduced to calculate the binding affinity of the generated molecules using the model-given conformations. This evaluation can more clearly identify the situations where the rationality of a generated 3D structure is poor, but it was scored with a high binding energy. For example, for the GraphBP results (shown in Table 1), although its docking affinity achieves the third-lowest outcome, its scoring affinity is surprisingly high, even being higher than zero, indicating that these pocket–ligand

conformations generated by GraphBP are thermodynamically unstable. By visualizing the output of GraphBP in Fig. 1, it can be found that the generated 3D structure is distorted and the ligand's center of mass dramatically deviates from the center of the pocket. Although a docking program can post-process GraphBP's design and partially resolve the problem by re-positioning the molecule in the center of the pocket, the irrationality of the molecular topology eventually results in a break up of the molecule into three fragments (see the docked pose of GraphBP in Fig. 1) during the conformational search. These fragments can more flexibly fill the interior of the pocket, resulting in a high estimated affinity. More detailed discussion of the delusion of docking energy could be found in Supplementary Section 5.

In comparison, SurfGen delivers the highest performance under both affinity prediction settings (docking and scoring), exceeding not

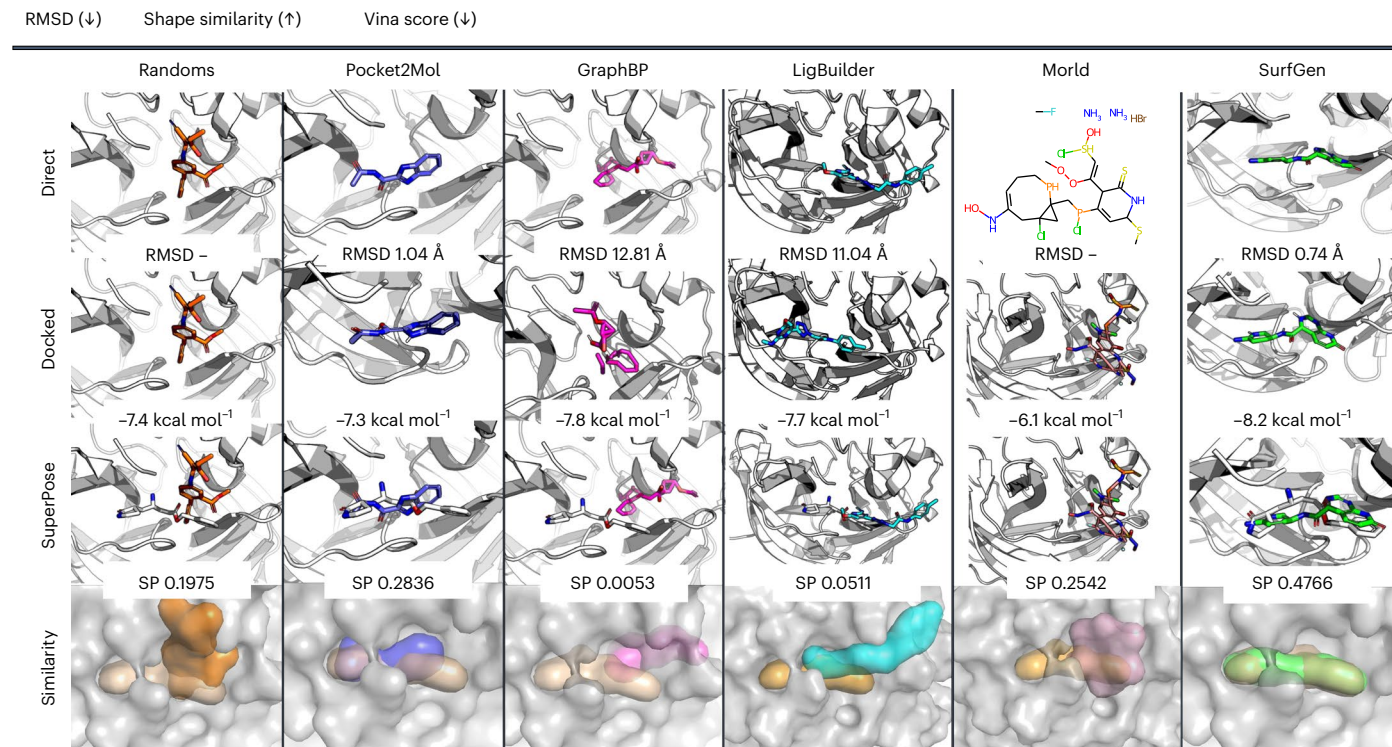


Fig. 1 | Generated molecules and randomly sampled molecules for a COVID-19 target, 3CL protein. Arrows pointing up or down indicate the optimal direction for the corresponding metrics. The first row denotes the directly generated conformations, but for randoms, only the docked conformation is accessible. The -F, HBr, and NH₃ in Morld are the broken fragments of the molecule after docking. The second row denotes the best docked conformations, and RMSD represents the root-mean-square deviation between the generated conformation

and docked structure. The value in kcal mol⁻¹ denotes the binding energy, and the binding affinity is estimated by AutoDock Vina. The third row shows the original ligand and direct generated molecules (for randoms, the docked result is shown). The fourth row shows the 3D shape similarity between the original ligands (light orange) and the generated molecules (orange, randoms; blue, Pocket2Mol; pink, GraphBP; cyan, LigBuilder; pink, Morld; green, SurfGen). SP denotes the shape similarity. The detailed definitions of RMSD and SP are given in Methods.

only the affinity of the original ligands by 3 kcal mol⁻¹ (-125-fold activity) but also that of the molecules generated by the other two SOTA deep learning methods, GraphBP and Pocket2Mol by about 1 kcal mol⁻¹ (-5-fold activity). This performance suggests that SurfGen has learned the energetically favorable distribution of the molecular conformations in the target pockets. The reinforcement learning-based method, Morld, which iterates molecules with the docking energy award, generates only the simplified molecular input line entry system (SMILES) string of molecules and thus lacks the scoring metric. The traditional approach, LigBuilder, competes closely with SurfGen, showing only 0.5 kcal mol⁻¹ higher binding energy. This finding fills in the blank in comparisons between classical and deep learning-based approaches, reminding us of the strength of classical methodology, that is, generating molecules by optimizing the given scoring function. However, the performance of LigBuilder on binding energy may reflect the methodological bias, as it generates molecules by optimizing the binding energy predicted by SCORE²¹ and applies the internal rules for molecule filtration. Conversely, SurfGen is an end-to-end approach, suggesting that the binding energy is not incorporated in the training and generation phase. Thus, the superior performance of SurfGen is from its learned topological and geometric interaction instead of the bias of a docking method, and other results, such as similarity to active molecules, provide further support for this claim.

We also evaluate the drug likeness of the generated compounds. As the drug-likeness assessment is a non-trivial task and current evaluation approaches are too crude, we believe that it is enough that the drug-likeness scores of the molecules generated by the model are close to those of the original ligands (that is, experimentally validated actives). LigBuilder achieves the highest SA and QED metric because it generates

molecules by assembling the predefined 57 fragments, inherently reducing molecular complexity. However, this strategy also limits the potential chemical space covered by LigBuilder, precluding the design of inhibitors that cannot be broken down into the stored fragments.

Another metric illustrating the superiority of our generated molecules is the molecular topological similarity. We calculate the similarity between the molecules generated by different methods and the molecules in the training and test sets, as well as their internal diversity. As shown in Supplementary Table 4, SurfGen achieves the highest similarity to the test and training sets, demonstrating its superior capability to learn models' distribution. Morld and LigBuilder were not trained on the CrossDock dataset, and thus there is no need to compute the similarity of them to the training and test molecules. The diversity metric is another critical metric in the evaluation of pocket-aware molecular generation, alongside binding energy. While high diversity may reduce the models' specificity to targets and low diversity may signify the mode collapse, SurfGen balances this aspect effectively, showing lower diversity than the random-like molecules generated by GraphBP but higher diversity than those generated by Morld. Of note, the results from Supplementary Table 4 are based on averaging over ~100 test targets, and the rationale of the discussions would be more obvious when introducing the negative control in the curated real-world datasets.

Drug design on COVID-19 and other therapeutic targets

While evaluating how a model performs on some representative test sets is a common practice in the machine learning community, scientists and drug designers are more concerned with these models' performances in real-world scenarios. To this end, we decide to further

Table 2 | The binding energies and drug-likeness properties of the top-5 molecules designed for three real-world targets

	Actives	Decoys	GraphBP	Pocket2Mol	Morld	LigBuilder	SurfGen
3CL top 5							
Docking energy (↓)	-7.599	-7.119	-7.479	-6.859	-6.180	-7.419	-8.540
QED	0.397	0.467	0.486	0.380	0.084	0.590	0.418
SA	0.696	0.703	0.7374	0.742	0.306	0.704	0.752
Lipinski	4.000	4.667	5.000	5.000	3.200	4.400	5.000
LogP	1.095	0.061	0.0976	-0.799	6.786	-0.297	-0.332
AKT1 top 5							
Docking energy (↓)	-11.940	-10.340	-9.059	-9.600	-8.020	-9.720	-12.62
QED	0.264	0.506	0.601	0.745	0.027	0.647	0.456
SA	0.266	0.314	0.454	0.264	0.300	0.712	0.422
Lipinski	4.200	4.800	5.000	5.000	2.400	5.000	5.000
LogP	2.475	1.208	1.417	0.967	9.252	1.720	1.748
CDK2 top 5							
Docking energy (↓)	-11.020	-9.920	-9.720	-9.300	-7.439	-9.660	-12.21
QED	0.407	0.563	0.581	0.667	0.178	0.646	0.748
SA	0.362	0.316	0.586	0.336	0.461	0.666	0.776
Lipinski	4.600	5.000	4.600	5.000	4.200	5.000	5.000
LogP	-0.893	0.821	1.765	0.397	4.510	1.658	0.756

All technical terms are discussed in the footnote of Table 1.

compare models on our curated benchmark containing more than 20 therapeutic targets (including COVID-19 targets) and their experimentally validated active ligands. We believe that this benchmark more realistically reflects these models' potential in a real-world setting. In this case, molecule generation is performed on these targets, and then binding affinity, drug-likeness properties and similarity with actives are evaluated. The difference between this experiment and the CrossDock test set experiment (in the previous section) is that there are 200 experimentally validated molecules collected for each target, whereas in CrossDock there is only one corresponding co-crystallized molecule for each target structure as a reference. Therefore, it is able to plot the activity distribution of the molecules experimentally proven to be active. In addition to collecting the experimentally active molecules as the positive controls, we also randomly select 200 molecules from the compound library as the negative controls.

Table 2 showcases the activity and corresponding drug-likeness properties of the top-5 molecules generated by different methods. In addition, we provide the visualizations of several SurfGen-generated molecules and the crystal molecule in Fig. 2b. Further taking the binding energy distributions shown in Fig. 2a and Supplementary Fig. 2 into account, a clear criterion for measuring a model's design capability emerges. Randomly selected molecules serve as the negative controls, representing a baseline for models that can generate feasible molecules but with no specificity toward the given targets. In contrast, experimentally validated active molecules serve as the positive controls, representing a benchmark for models capable of designing active-like molecules targeting specific proteins. The molecules generated by a SOTA model should ideally surpass the random distribution and closely approximate the active distribution, the more to the left the better. Furthermore, (1) the similarity to actives and (2) the diversity of the generated molecules for each target provides two additional supplementary views on the quality of the generative models (in addition to the perspective of the binding energy). A SOTA model should outperform the random baseline, otherwise, the model does not manifest a convincing sensitivity toward a given pocket. While the metrics of the output diversity operates similarly to the similarity metrics, the

actual consideration is more complex. Following the common principle that similar structures tend to exhibit similar effects, the active molecules targeting the same protein should thus share a certain degree of similarity^{24,25}. Consequently, the diversity metric for a SOTA method should be moderate, that is, lower than randomly selected molecules but still maintaining some diversity. While there is no rigorously pre-defined threshold, we argue that a top-performing model should outperform Morld on the diversity metric, as it generates molecules by iterating on the seed structure, thereby introducing correlations among the samples for each target. Thus, the diversity of these correlated molecules is minimal. In summary, the determination of a SOTA method rests on four criteria: (1) the binding energy distribution should be lower than that of random, the lower the better (illustrated in Fig. 2a and Supplementary Fig. 1); (2) the topological similarity to actives should be at least higher than that of randoms (results are shown in Table 3); (3) the diversity metric should be lower than randoms but higher than Morld (results are shown in Table 3); and (4) the 3D similarity metric, computed by the directly generated molecules against the crystal structure of the target, should be higher than the docked random molecules (results are shown in Supplementary Table 1).

As evidently shown in Fig. 2a, Tables 2 and 3, and Supplementary Table 1, GraphBP and Morld fail to fulfill any of the four SOTA criteria. The visualization in Fig. 1 shows that the plausibility of the molecules generated by these two models are low. Pocket2Mol, conversely, barely meets the similarity conditions 2 and 4 but shows a random-like binding energy distribution and diversity, implying room for improvement. The traditional method, LigBuilder, generates molecules with lower estimated binding energy than random, satisfying condition 1, and somewhat meets the two-dimensional similarity and diversity conditions 2 and 3. However, it fails to demonstrate superior performance on criterion 4, possibly due to the bias from its embedded binding energy method, SCORE, which is used for building molecules by optimizing a predefined scoring function. In contrast, SurfGen satisfies all four conditions, achieving SOTA performance on the real-world curated scenarios. Compared with GraphBP, the molecules generated by SurfGen not only have the lowest binding energy (even lower than

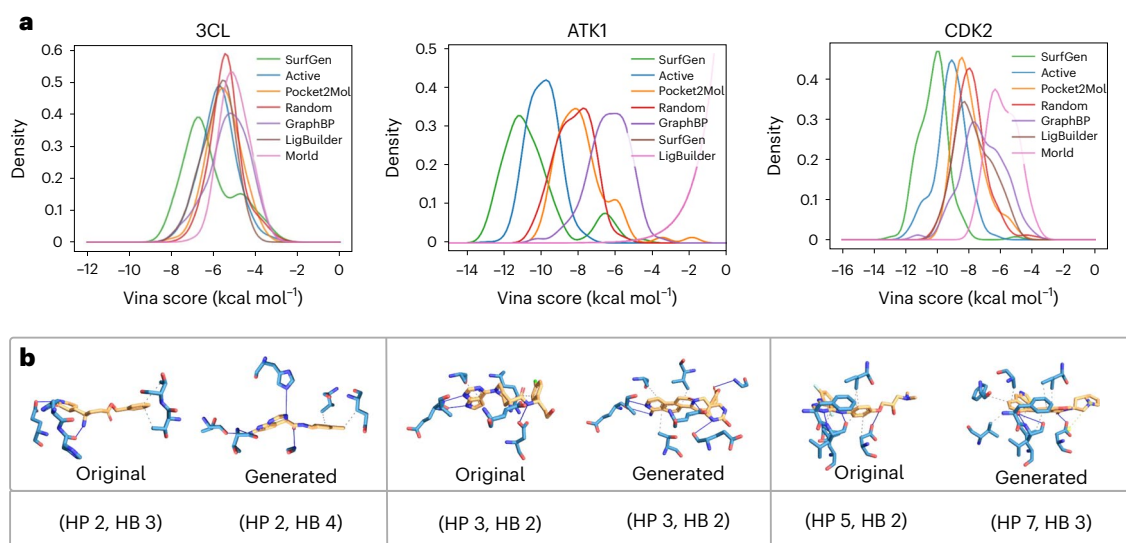


Fig. 2 | Illustration of SurfGen on real-world targets. a, The binding affinity distributions of the molecules generated by SurfGen, Pocket2Mol and GraphBP, experimentally active molecules (active) and randomly selected molecules (random). **b**, Examples of protein–ligand interaction analysis. HP denotes

hydrophobic interaction and HB denotes hydrogen-bonding interaction. Generated is the generated example and original is the original ligand that crystallizes with the protein.

Table 3 | Molecular similarity analysis on two real-world targets

Case	Random	GraphBP	Pocket2Mol	Morlid	LigBuilder	SurfGen
3CL (sim. act.)	0.2693 ± 0.11315	0.1558 ± 0.1073	0.1844 ± 0.0768	0.3968 ± 0.1735	0.2742 ± 0.1020	0.3213 ± 0.1518
Diversity (↑)	0.6891	0.8383	0.7984	0.0808	0.6969	0.6433
ATK1 (sim. act.)	0.1439 ± 0.1028	0.1568 ± 0.1008	0.1752 ± 0.1281	0.2021 ± 0.0796	0.3393 ± 0.0839	0.4583 ± 0.0861
Diversity (↑)	0.7331	0.8477	0.7455	0.1491	0.6897	0.3594
CDK2 (sim. act.)	0.2981 ± 0.0807	0.1573 ± 0.1087	0.2808 ± 0.087	0.3254 ± 0.1078	0.2787 ± 0.0969	0.5176 ± 0.1299
Diversity (↑)	0.7331	0.8361	0.7304	0.2416	0.7577	0.4361

Sim. act. is the molecular topological similarity between generated molecules and experimentally active molecules. Diversity is the internal similarity in each set of molecules.

those with experimental activity) but also present a high 3D overlap with the in situ ligand. In addition, the overlap results in a shape similarity score double that of the next best model, Pocket2Mol. Considering that SurfGen has no access to the active molecules in the training phase, it is believed that SurfGen has learned the accurate interaction within the pocket environments, designing molecules chemically and geometrically akin to active ones. Besides, unlike other structure-based molecule-generation methods^{26,27}, SurfGen does not require the guidance of seed molecules. It could be envisioned that for protein targets with known inhibitors, SurfGen could also be utilized to design potent ligand binders, which should be validated through bioassays in the future. For now, we provide the example of drug design through SurfGen on the COVID-19 target in the Git repository, providing experimentalists with more opportunities to accelerate COVID-19 drug discovery.

Mutants conditional molecular generation

As a pocket-based 3D molecule-generation model, SurfGen should be sensitive to subtle variations in the pocket structure, for example, generating hit compounds matching to different geometric and energetic profiles of closely related pockets, such as the pockets of wild-type and mutated targets. To verify SurfGen's sensitivity to these subtle changes in pocket structure, we conduct an experiment aiming to verify whether SurfGen can be an effective machine learning model to propose better drug candidates for mutated drug-resistant targets.

Mutations in some key residues of the pockets in protein targets lead to changes in their interactions with inhibitor molecules, one of the main causes of the off-target effect. To investigate the sensitivity of SurfGen to the mutations of some residues within the pocket, we designed the experiment for the shikimate protein (Protein Data Bank (PDB) ID 1ZYU ref. 28), which is a key target in the shikimate pathway, which is essential for the survival of *Mycobacterium tuberculosis* and an important antibacterial target²⁹.

The binding pattern of the shikimate kinase and shikimate is shown in Fig. 3a,b. The binding region of the native ligand shikimate is surrounded by three α -helices, and the key interactions of this system mainly include six residues, Asp34, Arg58, Glu61, Gly80, Gly81 and Arg136, as described in previous studies^{26,28}. This experiment is composed of two parts, multi-point mutation and single-point mutation. The multi-point mutation experiment consists of pocket (mutate all residues in native ligand 5 Å to alanine), interact (mutate all interacting key residues to alanine (Ala)) and pocket-no-interact (mutate all residues in native ligand 5 Å except the interacting key residues to Ala); whereas the single-point mutation experiment mutates all residues in the native ligand 5 Å to alanine in turn. Then SurfGen is utilized to generate molecules conditional on the wild type, multi-point mutants and single-point mutants. Examples are shown in Fig. 3c.

First, the molecular backbone generated on the wild-type structure is similar to that of the native ligand, as seen in Fig. 3c (wild type versus 1ZYU). Both share a hexatomic ring as the backbone structure and

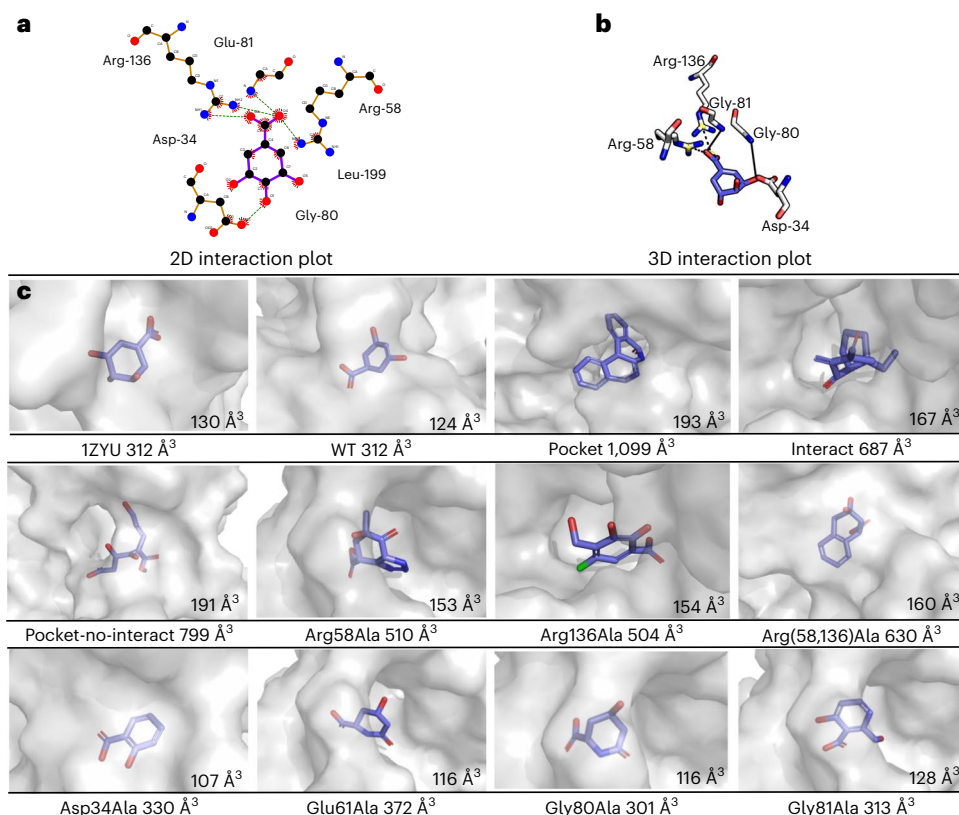


Fig. 3 | Conditional generated molecules on shikimate kinase mutants.

a, Two-dimensional interaction pattern for the wild-type crystal structure. The important residues are highlighted by the residue name ID. Carbon, black; oxygen, red; nitrogen, blue. **b**, Three-dimensional interaction pattern on the wild-type crystal structure. The important residues are also shown. Carbon,

purple; chlorine, green; oxygen, red; nitrogen, blue. **c**, Visualization examples of the generated samples toward the mutants. The volume at the bottom of image represents the volume of the molecules and the mutation notations are followed by the volume of the pocket.

occupy similar spatial positions and conformational angles in the protein pocket. The pocket mutation strategy in the multi-point mutation experiment produces a larger molecular structure that is more different from the native ligand. This is because all amino acid side-chain residues in the pocket are mutated to alanine, the smallest residue, resulting in a substantially larger protein pocket space ($312 \text{ \AA}^3 > 1,099 \text{ \AA}^3$), and leading SurfGen to generate a larger molecule ($130 \text{ \AA}^3 > 193 \text{ \AA}^3$) that is geometrically adapted to the enlarged mutated pocket. The interact mutation system on which six key residues are mutated to alanine generates a molecule that also differs from the native ligand. However, compared with the pocket with all pocket-residues mutated ($687 \text{ \AA}^3$ versus $1,099 \text{ \AA}^3$), the resulting molecule is smaller in size ($167 \text{ \AA}^3$ versus $193 \text{ \AA}^3$), as expected. The pocket-no-interact system mutates all residues in the pocket except for the key interaction residues. As illustrated in the pocket-no-interact (Fig. 3c), the backbone of the resulting molecule is similar to that of the ligand in the 1ZYU pocket, retaining the carboxyl groups involved in the interactions on the hexatomic ring and retaining more energy matching features, but with respect to geometry matching, the pocket structure is still much larger compared with the wild type ($312 \text{ \AA}^3 > 799 \text{ \AA}^3$) because of multiple mutated residues, which explains the production of a larger molecule ($130 \text{ \AA}^3 > 191 \text{ \AA}^3$).

Experiments with single-point mutations are discussed next. In both the Arg58Ala and Arg136Ala systems, we observe that the molecule generated by SurfGen still retains the ring backbone and the oxygen-containing group at the number one position, because the guanidine group in Arg58 and Arg136 can interact with that via the formation of hydrogen bonds or salt bridges. If these two key residues are mutated (as shown for Arg(58,136)Ala), the resulting molecule no longer carries a carboxyl group at the number one position of the hexatomic ring and the

rest of the backbone remains similar to that of the native ligand. A similar phenomenon is observed in the mutation experiment of Asp34Ala. As Asp34Ala mutates only asparagine at position 34, retaining the two arginines at positions 58 and 136, the resulting molecule maintains the carboxyl group that interacts with positions 58 and 136 and loses the hydroxyl group that interacts with Asp34. The remaining three mutant systems (Glu61Ala, Gly80Ala and Gly81Ala) produce molecules with smaller changes to the ring groups, probably because the residues at these three positions do not contribute as substantially to the formation of hydrogen bonds as residues such as arginine.

Discussion

SurfGen is a 3D molecular generation method inspired by the lock-and-key analogy. Although it shows strong design capabilities on both CrossDock benchmarks and real-world datasets, certain limitations remain. One notable issue is that SurfGen has only been evaluated on *holo* protein structures. These are structures where the protein pocket is already induced into a state conducive for ligand binding. The method has not yet been tested on *apo* structures, such as those predicted by AlphaFold. To make SurfGen adaptable for any given protein structure, an additional module for exploring protein conformational space should be considered—a possible avenue for future research. Another broader challenge pertains to the synthesizability of the generated molecules. Current molecular generation models are not yet sophisticated enough to produce optimal drug candidates directly. Medicinal chemists continue to have an irreplaceable role in artificial intelligence-based drug design, often needing to modify generated molecules to improve their synthesizability. Going forward, SurfGen aims to incorporate considerations around synthesizability

into the model's framework, thereby enhancing its intelligence across multiple target metrics.

Methods

As protein pockets are modeled as a channel of topological entity characterized by triangular surfaces in our approach, it presents obstacles in learning the underlying interactions directly using the existing 3D graph neural representation frameworks. As shown in Supplementary Fig. 1g, when aggregating information around atom i under the general Euclidean metric framework, the receptive field covers the back side of the pocket surface, but small molecules do not directly match with these backsides in reality. Therefore, we design Geodesic-GNN, a topological learning framework combined with equivariant constraint, to learn on curved surfaces with topological attention. In the Geodesic-GNN block, the perceptive field of atom i will not contain information of the back side, as shown in the bottom half of Supplementary Fig. 1g. Compared with the Euclidean geometric graph neural framework, Geodesic-GNN enables a proper scheme of information aggregation on the topological surface. For the interactions between ligand atoms and protein pockets, the spatial geometric interactions deserve more attention. Therefore, we design Geoattn-GNN based on the geometric attention mechanism for learning the matching laws between atoms in molecule and pockets. At this point, we have elucidated the roles of the two fundamental elements constituting SurfGen, and are ready to further introduce the representation learning framework. First, we map edge features and atomic features to a higher-dimensional representation space.

$$(s'_{n_i}, v'_{n_i}) = \text{GVP}(s_{n_i}^{l-1}, v_{n_i}^{l-1}) \quad (3)$$

$$(s'_{e_{ij}}, v'_{e_{ij}}) = \text{GVP}(s_{e_{ij}}^{l-1}, v_{e_{ij}}^{l-1}) \quad (4)$$

where $(s_{n_i}^{l-1}, v_{n_i}^{l-1})$ is the scalar and vector feature of node i in the $(l-1)$ th iteration and $(s_{e_{ij}}^{l-1}, v_{e_{ij}}^{l-1})$ is the scalar and vector feature of edge $i-j$ in the $(l-1)$ th iteration. GVP is the geometric vector perception³⁰. After that, for each point on the topological surface, Geodesic-GNN is performed to aggregate the information from its neighbors.

$$s_{m_i} = \text{Geodesic-GNN}((s_{n_i}, x_i, \hat{n}_i), (s_{n_j}, x_j, \hat{n}_j), (s_{e_{ij}}, v_{e_{ij}}, g_{ij})) \quad (5)$$

where x_i and $\hat{n}_i$ are the coordinate and normal vector of node i ; and g_{ij} is the geodesic distance between nodes i and j , subscript m denotes the message information. After aggregating the topological surface information, Geoattn-GNN is performed between the ligand atoms and the topological pockets, allowing SurfGen to learn the spatial interactions.

$$s_{m_i} = \text{Geoattn-GNN}((s_{n_i}, v_{n_i}), (s_{n_j}, v_{n_j}), (s_{e_{ij}}, v_{e_{ij}}, d_{ij})) \quad (6)$$

where d_{ij} is the Euclidean distance between nodes i and j . Then the learned interactions are exploited to guide the model selection and the assembly of atoms inside protein pockets for 3D molecular generation.

Geodesic convolutional block (Geodesic-GNN)

$$V_{n_i} = \{x_i, \hat{n}_i\} \quad (7)$$

$$s_{m_i} = \sum_j s_{n_j} \circ (\text{SL}(s_{e_{ij}})) \circ w(g_{ij}) \quad (8)$$

$$V_{m_i} = \sum_j \{ \text{VL}(V_{n_j}) \circ \text{SL}(s_{e_{ij}}) + \text{SL}(s_{n_j}) \circ \text{VL}(V_{e_{ij}}) \} \circ w(g_{ij}) \quad (9)$$

$$(s_{m_{ij}}^{(l)}, v_{m_{ij}}^{(l)}) = \text{GVP}(s_{m_{ij}}, v_{m_{ij}}) \quad (10)$$

where $\{x_i, \hat{n}_i\}$ is the vector features, position and normal vector of node i , g_{ij} is geodesic distance between i and j , s_{n_i} is node scalar feature of j , $s_{e_{ij}}$ is edge scalar feature of edge $i-j$, v_{n_i} and $v_{e_{ij}}$ are those of vector features. VL is the vector linear block¹⁴, SL is the scalar linear block and w is the active function, with the default being the sigmoid function. Output $s_{m_{ij}}^{(l)}, v_{m_{ij}}^{(l)}$ are edge message features.

Geometry attention convolutional block (Geoattn-GNN)

$$\alpha_{ij} = \text{LeakyRelu}(W \circ [s_{m_i}, \|s_{m_j}\|, d_{ij}]) \quad (11)$$

$$\alpha'_{ij} = \text{softmax}(\alpha_{ij}) = \frac{\exp(\alpha_{ij})}{\sum_j \exp(\alpha_{ij})} \quad (12)$$

$$A_{ij} = \text{tr}(V_{m_i} V_{m_j}^T) \quad (13)$$

$$A'_{ij} = \text{sigmoid}(A_{ij}) \quad (14)$$

$$s_{m_i} = \sum_j s_{n_j} \circ (\text{SL}(s_{e_{ij}})) \circ \alpha'_{ij} \quad (15)$$

$$V_{m_i} = \sum_j \{ \text{VL}(V_{n_j}) \circ \text{SL}(s_{e_{ij}}) + \text{SL}(s_{n_j}) \circ \text{VL}(V_{e_{ij}}) \} \circ A'_{ij} \quad (16)$$

$$(s_{m_{ij}}^{(l)}, v_{m_{ij}}^{(l)}) = \text{GVP}(s_{m_{ij}}, v_{m_{ij}}) \quad (17)$$

where (α'_{ij}, A'_{ij}) is the scalar and vector attention score between nodes i and j ; $\parallel$ is the concatenate operator; and $\text{tr}()$ is the trace operator.

Energy and geometry features on the pocket surface

Three geometric and three energetic features are extracted and attached to the nodes of the pocket surface, where the geometric features include Gaussian curvature, mean curvature and shape index; the energetic features include Poisson-Boltzmann charge³¹, hydrogen-bond donors³² and hydrophobicity³³. The detailed descriptions could be found in Supplementary Section 7.

Generation module

The molecule-generation methods can be categorized into autoregressive^{16,18,34} and one-shot^{15,23,35} schemes. The one-shot generation methods generally require a pre-predicted molecular matrix, that is, the number of atoms is predicted first, and the atomic classes, position coordinates and bonding relationships are determined together in the following step. In SurfGen, the autoregressive protocol is adopted so that the generated atoms can fully interact with pocket environment and thus learn geometry matching more efficiently.

Focal atom location. (When the pocket is empty, that is, $t = 0$.) After Geodesic aggregation on the surface of the protein channel, SurfGen is supposed to select the starting node from the nodes of the topological surface. That is, a network is trained to predict the starting probability for each node on the surface.

$$(s_f, v_f) = \text{GVP}(s_h, v_h) \quad (18)$$

$$p_f = \sigma(\text{MLP}_{\text{foc}_2}(\text{MLP}_{\text{foc}_1}(s_f) + \|v_f\|_2)) \quad (19)$$

where (s_h, v_h) is the scalar and vector features in hidden space, and σ is the sigmoid active function, which outputs a probability value at $[0, 1]$.

When there are already generated atoms inside the protein pocket, the focal atom was selected from the already generated atoms. The formula is the same as the circumstance of $t = 0$.

Geometry generation. To satisfy the equivariance requirement of the geometry generation, that is, predict the coordinate of the next atom, the techniques from VP¹⁴ and GVP³⁰ were adopted here to construct this module. In addition, the negative log likelihood (NLL) loss function was used to fit a 3D multivariate Gaussian distribution:

$$p(\Delta r_i) = \sum_{k=1}^K \pi_i^{(k)} \mathcal{N}(\mu_i^{(k)}, \Sigma_i^{(k)}) \quad (20)$$

$$p(\Delta r_i + r_{f_i}) = \sum_{k=1}^K \pi_i^{(k)} \mathcal{N}(\mu_i^{(k)} + r_{f_i}, \Sigma_i^{(k)}) \quad (21)$$

$$(s_\pi, V_\pi) = \text{GVP}_\pi(s_h, V_h) \quad (22)$$

$$(s_\mu, V_\mu) = \text{GVP}_\mu(s_h, V_h) \quad (23)$$

$$(s_\Sigma, V_\Sigma) = \text{GVP}_\Sigma(s_h, V_h) \quad (24)$$

$$\mu = \mu_v \quad (25)$$

$$\Sigma = \text{MLP}_\Sigma(s_\Sigma) \quad (26)$$

$$\pi = \text{MLP}_{\pi_2}(\text{MLP}_{\pi_1}(s_\pi) + \|V_\pi\|_2) \quad (27)$$

where r_i is the Cartesian coordinate of i th generated atom; $\mathcal{N}$ is the Gaussian distribution, π , μ , Σ , and K are the combination coefficient, mean, variance, and the number of components of the Gaussian mixed density network; (s_h, V_h) is the scalar and vector feature of hidden interaction space.

Topology generation. Knowing the spatial relationship between the next atom and other existing atoms, the atomic type and bonding relationship of the new atom can be determined. However, considering that the distribution of atom types inside the pocket is influenced by the chemical environment, SurfGen utilizes the structural k nearest neighbor (kNN) aggregation mechanism in the topology-generation module.

$$(s_{m(i)}, V_{m(i)}) = \sum_{k \in \text{kNN}(i)} \text{Message}(s_{h(k)}, V_{h(k)}, e_{ki}, U_{ki}) \quad (28)$$

$$p_{a_i} = \text{MLP}_{a_2}(\text{MLP}_{a_1}(s_{m(i)}) + \|V_{m(i)}\|_2) \quad (29)$$

$$e'_{ij}, U'_{ij} = \text{GVP}_{bp_1}([s_i \| s_j \| e_{ij}], [V_i \| V_j \| U_{ij}]) \quad (30)$$

$$e''_{ij}, U''_{ij} = \text{Attention}(e'_{ij}, U'_{ij}) + (e'_{ij}, U'_{ij}) \quad (31)$$

$$e_{ij}^{\text{type}} = \text{MLP}_{bp_2}(\text{MLP}_{bp_1}(e''_{ij}) + \|U''_{ij}\|_2) \quad (32)$$

where Message is the message-passing framework; $(s_{h(k)}, V_{h(k)}, e_{ki}, U_{ki})$ are scalar and vector features of atoms and edges; p_{a_i} is the probability of types of next atom; Attention^{36,37} is triangle attention module; and e_{ij}^{type} is the bonding relationship between the next atom and other atoms.

Loss function

The model's training objectives are intrinsically linked to the generation process. To articulate, each component of the generation process is associated with a specific loss function to optimize during training.

More specifically, the focal atom prediction module undergoes training with the binary cross-entropy loss. The module that generates geometric aspects is trained using the negative log-likelihood loss. Finally, the topology-generation module, which predicts atomic type and bonding relationships, is trained employing the cross-entropy loss.

Other set-ups

CrossDock2020 dataset. Following previous work, SurfGen is trained on the CrossDock2020 dataset²², which contains 22.5 million protein–ligand pairs. Similar to the set-up of SBDD³⁸ and LiGAN¹⁷, we use the docked conformations within 2 Å of the crystal conformation as the training data.

Real-world therapeutic targets. Protein structures are taken from the PDB³⁹, while the experimentally validated actives are curated from BindingDB⁴⁰. The target of the COVID-19 case is 3CL protease, which is essential across a broad spectrum of coronaviruses, and the PDB ID is 6LU7 ref. 41.

RMSD. RMSD is the root-mean-square deviation, which is used to measure the difference between conformations of the same molecule. The smaller the value of RMSD, the closer the conformation of two molecules is. The formula for calculating RMSD is as follows:

$$\text{RMSD}(\bar{R}, R) = \left(\frac{1}{n} \sum_{i=1}^n \|\bar{R}_i - R_i\|^2 \right)^{\frac{1}{2}} \quad (33)$$

where n is the number of atoms; R is the molecular conformation; and R_i is the i th atomic Cartesian coordinate. Figure 1 shows the RMSD of the directly generated conformation by the model and the optimal conformation after docking.

Molecular similarity. The molecular similarity, also referred to as molecular topological similarity (Table 2 and Supplementary Table 4) is the Tanimoto coefficient⁴² between different molecular fingerprints. The Tanimoto coefficient is defined as follows.

$$\text{Tanimoto}(G_1, G_2) = \frac{|G_1 \cap G_2|}{|G_1 \cup G_2|} = \frac{|G_1 \cap G_2|}{|G_1| + |G_2| - |G_1 \cap G_2|} \quad (34)$$

where the G is the molecular fingerprint. In this work, we use default topological fingerprints in RDKit⁴³.

Docking method. AutoDock Vina²⁰ is used in the assessments of generated molecules. Note, the docking method can provide only the static binding energy, while the kinetic aspect is also another important factor for protein–ligand binding. However, computing the kinetic properties of ligands targeting the protein, such as residence time, is non-trivial and task specific. In addition, it often requires long-time dynamical simulation. Thus the docking serves as the major tool for estimating the ligand's binding capability.

Concerns of overfitting. To mitigate the risk of overfitting, several precautionary measures have been implemented. First, the model's performance is evaluated using a well-established benchmark, the CrossDock dataset. In addition, the model is subjected to further validation on a curated dataset that simulates real-world scenarios. Moreover, the early stopping technique is employed to determine the model parameters, which are selected based on achieving the lowest validation loss.

Statistics and reproducibility

No statistical method was used to predetermine sample size.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data are available at Zenodo (<https://doi.org/10.5281/zenodo.8307911>)⁴⁴. PDB IDs 1ZYU and 6LU7 are available in the PDB (<https://www.rcsb.org/>). Source data are available with this paper.

Code availability

The code is available at GitHub (<https://github.com/HaotianZhangAI4Science/SurfGen>)⁴⁵.

References

1. Ferreira, L. G., Dos Santos, R. N., Oliva, G. & Andricopulo, A. D. Molecular docking and structure-based drug design strategies. *Molecules* **20**, 13384–13421 (2015).
2. Anderson, A. C. The process of structure-based drug design. *Chem. Biol.* **10**, 787–797 (2003).
3. Shoichet, B. K. Virtual screening of chemical libraries. *Nature* **432**, 862–865 (2004).
4. Böhm, H.-J. The computer program LUDI: a new method for the de novo design of enzyme inhibitors. *J. Comput. Aided Mol. Des.* **6**, 61–78 (1992).
5. Wang, R., Gao, Y. & Lai, L. LigBuilder: a multi-purpose program for structure-based drug design. *Mol. Model. Annu.* **6**, 498–516 (2000).
6. David, L., Nielsen, P. A., Hedstrom, M. & Norden, B. Scope and limitation of ligand docking: methods, scoring functions and protein targets. *Curr. Comput. Aided Drug Design* **1**, 275–306 (2005).
7. Jorgensen, W. L. Rusting of the lock and key model for protein-ligand binding. *Science* **254**, 954–955 (1991).
8. Ain, Q. U., Aleksandrova, A., Roessler, F. D. & Ballester, P. J. Machine-learning scoring functions to improve structure-based binding affinity prediction and virtual screening. *Wiley Interdiscip. Rev. Comput. Mol. Sci.* **5**, 405–424 (2015).
9. McNutt, A. T. et al. GNINA 1.0: molecular docking with deep learning. *J. Cheminformatics* **13**, 1–20 (2021).
10. Shen, C. et al. Boosting protein–ligand binding pose prediction and virtual screening based on residue–atom distance likelihood potential and graph transformer. *J. Med. Chem.* **65**, 10691–10706 (2022).
11. Jiang, D. et al. Interactiongraphnet: a novel and efficient deep graph representation learning framework for accurate protein–ligand interaction predictions. *J. Med. Chem.* **64**, 18209–18232 (2021).
12. Moon, S., Zhung, W., Yang, S., Lim, J. & Kim, W. Y. PIGNet: a physics-informed deep learning model toward generalized drug–target interaction predictions. *Chem. Sci.* **13**, 3661–3673 (2022).
13. Gainza, P. et al. Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning. *Nat. Methods* **17**, 184–192 (2020).
14. Deng, C. et al. Vector neurons: A general framework for so (3)-equivariant networks. *Proc. IEEE/CVF International Conference on Computer Vision* 12200–12209. (2021).
15. Zang, C. & Wang, F. Moflow: an invertible flow model for generating molecular graphs. *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. 617–626 (2020).
16. Peng, X. et al. Pocket2Mol: efficient molecular sampling based on 3D protein pockets. *International Conference on Machine Learning*. 17644–17655. (2022).
17. Ragoza, M., Masuda, T. & Koes, D. R. Generating 3D molecules conditional on receptor binding sites with deep generative models. *Chem. Sci.* **13**, 2701–2713 (2022).
18. Liu, M., Luo, Y., Uchino, K., Maruhashi, K. & Ji, S. Generating 3D molecules for target protein binding. *International Conference on Machine Learning*, 13912–13924. (2022).
19. Jeon, W. & Kim, D. Autonomous molecule generation using reinforcement learning and docking to develop potential novel inhibitors. *Sci. Rep.* **10**, 22104 (2020).
20. Trott, O. & Olson, A. J. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J. Comput. Chem.* **31**, 455–461 (2010).
21. Wang, R., Liu, L., Lai, L. & Tang, Y. SCORE: a new empirical method for estimating the binding affinity of a protein–ligand complex. *Mol. Model. Annu.* **4**, 379–394 (1998).
22. Francoeur, P. G. et al. Three-dimensional convolutional neural networks and a cross-docked data set for structure-based drug design. *J. Chem. Inf. Model.* **60**, 4200–4215 (2020).
23. Schneuing, A. et al. Structure-based drug design with equivariant diffusion models. Preprint at <https://arxiv.org/abs/2210.13695> (2022).
24. Martin, Y. C., Kofron, J. L. & Traphagen, L. M. Do structurally similar molecules have similar biological activity? *J. Med. Chem.* **45**, 4350–4358 (2002).
25. Yang, J., Cai, Y., Zhao, K., Xie, H. & Chen, X. Concepts and applications of chemical fingerprint for hit and lead screening. *Drug Discov. Today* **27**, 103356 (2022).
26. Kang, S.-G. et al. In-pocket 3D graphs enhance ligand–target compatibility in generative small-molecule creation. Preprint at <https://arxiv.org/abs/2204.02513> (2022).
27. Wang, M. et al. Relation: a deep generative model for structure-based de novo drug design. *J. Med. Chem.* **65**, 9478–9492 (2022).
28. Gan, J., Gu, Y., Li, Y., Yan, H. & Ji, X. Crystal structure of *Mycobacterium tuberculosis* shikimate kinase in complex with shikimic acid and an ATP analogue. *Biochemistry* **45**, 8539–8545 (2006).
29. Pereira, J. H. et al. Shikimate kinase: a potential target for development of novel antitubercular agents. *Curr. Drug Targets* **8**, 459–468 (2007).
30. Jing, B., Eismann, S., Suriana, P., Townshend, R. J. & Dror, R. Learning from protein structure with geometric vector perceptrons. Preprint at <https://arxiv.org/abs/2009.01411> (2020).
31. Lamm, G. The Poisson–Boltzmann equation. *Rev. Comput. Chem.* **19**, 147–365 (2003).
32. Kortemme, T., Morozov, A. V. & Baker, D. An orientation-dependent hydrogen bonding potential improves prediction of specificity and structure for proteins and protein–protein complexes. *J. Mol. Biol.* **326**, 1239–1259 (2003).
33. Hagemans, D., Van Belzen, I. A., Morán Luengo, T. & Rüdiger, S. G. A script to highlight hydrophobicity and charge on protein surfaces. *Front. Mol. Biosci.* **2**, 56 (2015).
34. Shi, C. et al. Graphaf: a flow-based autoregressive model for molecular graph generation. *International Conference on Learning Representations (ICLR)*, 2020.
35. Lin, H. et al. DiffBP: generative diffusion of 3D molecules for target protein binding. Preprint at <https://arxiv.org/abs/2211.11214> (2022).
36. Jumper, J. et al. Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
37. Lu, W. et al. TANKBind: Trigonometry-aware neural networks for drug–protein binding structure prediction. *Adv. Neural Inf. Process. Syst.* **35**, 7236–7249 (2022).
38. Luo, S., Guan, J., Ma, J. & Peng, J. A 3D generative model for structure-based drug design. *Adv. Neural Inf. Process. Syst.* **34**, 6229–6239 (2021).
39. Burley, S. K. et al. Protein Data Bank (PDB): the single global macromolecular structure archive. *Protein Crystallogr.* **1607**, 627–641 (2017).
40. Liu, T., Lin, Y., Wen, X., Jorissen, R. N. & Gilson, M. K. BindingDB: a web-accessible database of experimentally determined protein–ligand binding affinities. *Nucleic Acids Res.* **35**, D198–D201 (2007).

41. Jin, Z. et al. Structure of M^{pro} from SARS-CoV-2 and discovery of its inhibitors. *Nature* **582**, 289–293 (2020).
42. Tanimoto, T. T. *Elementary Mathematical Theory of Classification and Prediction*, IBM Internal Report (1958).
43. Landrum, G. RDKit documentation. *Release* **1**, 4 (2013).
44. Odi, n Z. CrossDock processed data. *Zenodo* <https://doi.org/10.5281/zenodo.7751348> (2023).
45. Odin, Z. SurfGenV1. *Zenodo* <https://doi.org/10.5281/zenodo.8307911> (2023).
46. Clark, D. E. & Pickett, S. D. Computational methods for the prediction of ‘drug-likeness’. *Drug Discov. Today* **5**, 49–58 (2000).
47. Ertl, P. & Schuffenhauer, A. Estimation of synthetic accessibility score of drug-like molecules based on molecular complexity and fragment contributions. *J. Cheminformatics* **1**, 1–11 (2009).
48. Ganesan, A. The impact of natural products upon modern drug discovery. *Curr. Opin. Chem. Biol.* **12**, 306–317 (2008).
49. Sangster, J. Octanol–water partition coefficients of simple organic compounds. *J. Phys. Chem. Ref. Data* **18**, 1111–1229 (1989).

Acknowledgements

This work was financially supported by the National Key Research and Development Program of China (2022YFF1203003), the National Natural Science Foundation of China (22220102001, 82373791, and 81973281) and the Natural Science Foundation of Zhejiang Province (LD22H300001).

Author contributions

O.Z. contributed to the main idea and code. T.W. and N.W. contributed to the paper writing and code reorganization. G.W. contributed to the collection of the dataset and the corresponding experiment. D.J. contributed to the real-world case of the COVID-19 target experiment. X.W., H.Z. and J.W. contributed to the data analysis and drawing. N.W. contributed to the assessment of LigBuilder and Morl methods. E.W. contributed to the instruction in physical concepts. G.C. and Y.D. contributed to the visualization and technique support. P.P. contributed to the suggestion of the mutation experiment with

molecular generation protocol. Y.K. and C.-Y.H. contributed to the paper revision and experimental design. T.H. contributed to the essential financial support and conception, and was responsible for the overall quality.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s43588-023-00530-2>.

Correspondence and requests for materials should be addressed to Yu Kang, Chang-Yu Hsieh or Tingjun Hou.

Peer review information *Nature Computational Science* thanks Huziel Saucedo and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Kaitlin McCardle, in collaboration with the *Nature Computational Science* team.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature America, Inc. 2023

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☒ ☐ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☒ ☐ A description of all covariates tested
- ☒ ☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☒ ☐ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☒ ☐ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

- python 3.6
- pandas '1.1.5'; numpy '1.18.4';

Data analysis

- pandas '1.1.5'; numpy '1.18.4'; torch '1.19.0+cu113'; qVina02; PLIP '0.0.1'; rdkit '2020.09.1.0'; pyvol '1.7.6'; pymol '2.5.0'; matplotlib '3.7.0';
LigPlot '4.5.3'
Code in the original research
<https://github.com/HaotianZhangAI4Science/SurfGen/tree/main>

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The data is available at Zenodo (<https://doi.org/10.5281/zenodo.8307911>).

PDB IDs - 1zyu, 6lu7 are available in PDB (<https://www.rcsb.org/>).

Source data for Figure 1-3 and Table 1-3 are available with this manuscript.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender

N/A

Population characteristics

N/A

Recruitment

N/A

Ethics oversight

N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

The data size of the benchmark is CrossDock with 100,000 protein-ligand pairs, 18 real-world pharmaceutic targets with for corresponding 200 molecules each target. The reason for evaluating on CrossDock because it is an established benchmark used for other methods. Plus, it is the largest dataset for training 3D structure-aware molecular generation models as far as we know. The test/train split is done in terms of 40% sequence similarity according to its original report.

Data exclusions

Only the data that cannot be parsed by rdkit are excluded from analysis. This is a generally acceptable approach for peer methods.

Replication

We replicated the experiments three times. The attempts to verify the reproducibility of the results are successful.

Randomization

The training set and the test set were splitted according to the 40% sequence similarity according to the dataset original reports. The random seeds of the models are also fixed.

Blinding

The blinding was not relevant to our study since this work is not related with the wet experiments.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging